

Herding cats: Using the Big Five to organise reinforcement sensitivity theory scales

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Abstract

Over the past 40 years, the Reinforcement Sensitivity Theory (RST) of personality has inspired a proliferation of self-report scales, potentially creating ‘jingle-jangle’ fallacies and a fragmented empirical literature. The Big Five personality taxonomy may offer a useful organising framework for these scales, as it seems plausible that many could be represented as narrow facets of the broader Big Five domains. Across two studies, we explored (Study 1, $N = 408$) then attempted to confirm (Study 2, $N = 423$) whether and where a collection of >25 RST scales might be located within the space of the Big Five. Both studies revealed that more than 70% of these RST scales were more strongly related to the Big Five domains than more than 10% (i.e., 3/30) of Big Five facets, suggesting that they too could plausibly be regarded as Big Five facets. Moreover, several RST scales were highly correlated ($r > .80$) with individual Big Five facets, indicating potential redundancy. These results suggest that the Big Five can indeed provide a useful organising framework for most RST scales, which may help researchers grappling with construct proliferation and jingle-jangle fallacies within the RST literature.

Keywords

reinforcement sensitivity theory, Big Five, construct proliferation, jingle-jangle fallacies, personality structure

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“The RST field is becoming increasingly muddled – an unwelcome state of affairs because it is bound to impede the scientific progress of RST.”

Phillip J. Corr (2016), p. 60

Personality psychology—and other areas of psychology that rely on psychometric scales—is plagued by two problems. The first is *construct proliferation*, whereby new constructs and/or scales to measure them are continually introduced into the literature (Bainbridge et al., 2022). The second problem, exacerbated by the first, is *jingle-jangle fallacies* (Pedhazur & Schmelkin, 1991), whereby scales with the same name are presumed to measure the same construct (jingle), and scales with different names are presumed to measure different constructs (jangle). One area where both problems are steadily growing is Reinforcement Sensitivity Theory (RST; Corr,

2008). Several questionnaires have been developed to measure the constructs specified within RST, based both on the original version of the theory (o-RST; Gray, 1970, 1973) as well as its subsequent revision (r-RST; Gray & McNaughton, 2000; McNaughton & Corr, 2004). Indeed, there are now around a dozen RST questionnaires in circulation, each reflecting different conceptualisations and operationalisations of the traits specified within the theory (i.e., there is construct/scale proliferation). It is unclear whether various similarly-named RST scales

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converge (i.e., there are potential jingle fallacies), whether differently-named RST scales diverge (i.e., there are potential jangle fallacies), or which, if any, faithfully reflect their intended constructs.

The current state of RST-inspired psychometric assessment is reminiscent of the situation that once characterised personality psychology as a whole. Specifically, the mid-20th Century was marked by “an explosion of personality inventories” (Revelle et al., 2011, p. 9), and for a while “the only good way to organise the plethora of possible personality traits was that provided by the alphabet” (McAdams, 1997, p. 28). This situation was greatly improved by the discovery of the Big Five (Goldberg, 1990) and similar models of personality trait structure (most notably the HEXACO, Ashton & Lee, 2001) offering parsimonious sets of dimensions for navigating trait space. Recently, Bainbridge et al. (2022) found that the Big Five provided a useful organising framework for the vast majority of ~40 highly-cited psychological trait scales. Many of these scales could be plausibly represented as facets of the Big Five—indeed, several appeared redundant with existing facets. These results showed how trait taxonomies can help to make sense of the similarities and differences among various scales by locating them within a common framework. The Big Five may similarly provide a useful framework for the plethora of RST scales that have been developed over the last three decades.

In the following sections, we briefly outline the details of o-RST and r-RST, its presumed links with personality traits, and some of the most prominent RST scales. We then explain how the Big Five can be evaluated as a potential organising framework for these scales, and the benefits that may result from doing so.

A brief overview of reinforcement sensitivity theory

RST is a theory of the biological bases of personality, first proposed by Gray (1970, 1973, 1982), and subsequently revised by Gray and McNaughton (2000).¹ The history and development of the theory, along with its specific neurobiological details, are mostly irrelevant to the aims of our study, which focuses squarely on questionnaires inspired by RST. Thus, our brief overview of RST will focus only on details necessary for understanding the design of the various RST questionnaires described later (for more in-depth summaries of the theory itself, interested readers should consult Corr, 2004; Pickering & Corr, 2008; Smillie et al., 2006).

At the heart of RST is Gray’s (1973, 1977, 1987) proposal that three biological systems regulate responses to rewards and punishments, and that individual

differences in the functioning of these systems manifest as stable personality traits. The three systems described in o-RST were (1) a *Behavioural Inhibition System* (BIS), activated by conditioned punishment signals, and producing inhibition / passive avoidance, and feelings of anxiety; (2) a *Fight-Flight System* (FFS), activated by unconditioned punishment signals, producing defensive aggression (fight), escape (flight), and feelings of fear / panic; and (3) a *Behavioural Activation System* (or *Behavioural Approach System*, BAS), activated by conditioned reward signals, producing reward-approach behaviour and feelings of desire / enthusiasm. In terms of personality, Gray (1981, 1987) suggested that BIS-sensitivity manifests as trait anxiety, whereas BAS-sensitivity manifests as trait impulsivity (o-RST did not clearly link FFS with personality). Drawing on the most salient model of personality structure at the time, Gray (1981, 1987) represented anxiety and impulsivity as a modification to Eysenck’s (1967) two-dimensional model of personality structure, rotating neuroticism anti-clockwise to form anxiety and extraversion anti-clockwise to form impulsivity (see Figure 1).

The revision to Gray’s theory, r-RST (Gray & McNaughton, 2000; McNaughton & Corr, 2004), was prompted by three decades of data informing refinements to each of the biological systems: (1) The BAS was now understood to be activated by all signals of reward (not just conditioned); (2) the FFS was now understood to be activated by all signals of punishment, and to also produce ‘freezing’ responses to inescapable threat (and was thus relabelled as the FFFS); and, most distinctively, (3) the BIS was now understood to be activated by goal conflict, the most salient example of which is an approach-avoidance conflict, elicited by incompatible signals of reward and punishment. Among other clarifications, r-RST offered a sharper distinction between

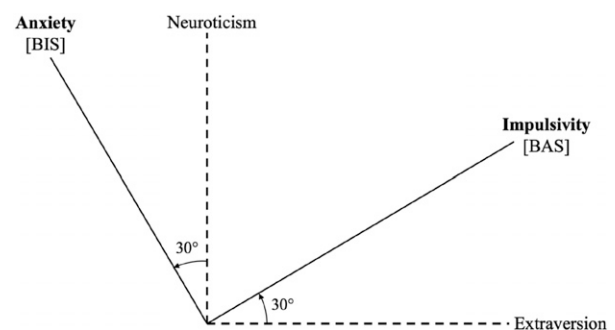


Figure 1. The putative location of anxiety (BIS-sensitivity) and Impulsivity (BAS-sensitivity) within the space of Eysenck’s (1967) Neuroticism-Extraversion model. *Note.* Adapted from Corr (2004); There has been disagreement as to whether a 30° or 45° is most appropriate (see Pickering et al., 1999; Smillie et al., 2006).

fear and anxiety, described in terms of ‘defensive direction’ (McNaughton & Corr, 2004). Specifically, in response to punishments or threats, the FFFS was held to generate feelings of fear and ‘defensive avoidance’ of the fear-inducing stimuli by either escaping (flight), freezing if escape is not possible, or engaging in defensive attack (fighting) if the threat is extremely close. Conversely, in a goal-conflict situation characterised by signals of both reward and punishment, the BIS was understood to be engaged by joint activation of the BAS and the FFFS (see Smillie et al., 2006, Figure 2), generating feelings of anxiety and ‘defensive approach’, comprising risk assessment behaviours directed toward the potentially threatening stimulus (Corr, 2004; Perkins et al., 2012).

In terms of links with personality traits, r-RST did not entail an explicit revision to the scheme depicted in Figure 1. However, it did imply a shift in how RST-related traits are to be conceptualised (see Smillie et al., 2006, for detailed discussion). For example, whereas BIS-sensitivity/anxiety based on o-RST was conceptualised in terms of one’s responses to imminent punishments or threats (e.g., being approached by a snarling dog), under r-RST it was conceptualised in terms of one’s responses to conflicting stimuli or incompatible goals (e.g., walking through a rough part of town to get to the safety of your car). Furthermore, both o-RST and r-RST offered fairly specialized conceptualisations of anxiety and impulsivity, leading many to suggest that existing measures of these traits were

unsuitable for testing RST (e.g., Carver & White, 1994; Torrubia et al., 2001). For this reason, researchers sought to develop purpose-built RST scales.

The proliferation of reinforcement sensitivity theory scales

Over the last 40 years, several attempts have been made to develop scales for measuring individual differences in BIS, BAS, and FFFS sensitivity (for a review see Corr, 2016). RST has been described as a complex theory that “often masquerades as simple one” (Pickering, 2008, p. 455), which may account for the sheer number and variety of such scales that have spilled into the literature. The first wave of scales, inspired by the o-RST, were followed by a second wave reflecting r-RST (see Table 1), amid many attempts to clarify and revise these measures (e.g., Corr, 2001; Smillie et al., 2006). As Corr (2016) describes, RST researchers are now “faced with a large (and somewhat bewildering) diversity of questionnaires from which to choose” (p. 60). We now describe four prominent examples, two based on o-RST—the BIS/BAS Scales (Carver & White, 1994) and the Sensitivity to Reward and Punishment Questionnaire (SPSRQ; Conner et al., 2018; Torrubia et al., 2001)—and two based on r-RST questionnaires—the Jackson-5 (Jackson, 2009) and the RST Personality Questionnaire (RST-PQ; Corr & Cooper, 2016).

Table 1. RST questionnaires inspired by o-RST and r-RST.

Theory	Questionnaire	Citation	Scales		
			BIS	BAS	FFS/FFFS
o-RST	SP	Torrubia and Tobena (1984)	✓ SP	×	×
	GWPPQ	Wilson et al. (1989)	✓ BIS: PA, Ext	✓ BAS: Ap, AA	✓ FFS: Fi, FI
	GRAPES	Ball and Zuckerman (1990)	✓ RE	✓ PE	×
	BIS	MacAndrew and Steele (1991)	✓ BIS	×	×
	BIS/BAS	Carver and White (1994)	✓ BIS	✓ BAS: RResponse, Dr, FS	×
r-RST	SPSRQ	Torrubia et al. (2001)	✓ SP	✓ SR	×
	Jackson-5	Jackson (2009)	✓ BIS	✓ BAS	✓ FFFS: Fi, FI, Fr
	RSQ	Smederevac et al. (2014)	✓ BIS	✓ BAS	✓ FFFS: Fi, FI, Fr
	rRST-Q	Reuter et al. (2015)	✓ BIS	✓ BAS	✓ FFFS: Fi, FI, Fr
	RST-PQ	Corr and Cooper (2016)	✓ BIS: MPI, BD, OT, CRA	✓ BAS: RI, GDP, RReact, Imp	✓ FFA: Fi, FI, AA
	RST-AQ	Franchina et al. (2024)	✓ BIS	✓ BAS: J, Int, Act	✓ FFFS: Fi, FI, Fr

Note. o-RST = Original Reinforcement Sensitivity Theory; r-RST = Revised Reinforcement Sensitivity Theory; SP = Sensitivity to Punishment Scale; GWPPQ = Gray-Wilson Personality Questionnaire; GRAPES = General Reward and Punishment Expectancy Scales; BIS = BIS Scale; BIS/BAS = BIS/BAS Scales; SPSRQ = Sensitivity to Punishment, Sensitivity to Reward Questionnaire; Jackson-5 = Jackson-5 Scales; RSQ = Reinforcement Sensitivity Questionnaire; rRST-Q = Revised Reinforcement Sensitivity Theory Questionnaire; RST-PQ = Reinforcement Sensitivity Theory Personality Questionnaire; RST-AQ = Reinforcement Sensitivity Theory Affects Questionnaire; BIS = Behavioural Inhibition System; PA = Passive Avoidance; Ext = Extinction; BAS = Behavioural Activation System; Ap = Approach; AA = Active Avoidance; FFS = Fight-Flight System; Fi = Fight; FI = Flight; RE = Reward Expectancy; PE = Punishment Expectancy; RRes = Reward Responsiveness; Dr = Drive; FS = Fun Seeking; SR = Sensitivity to Reward; FFFS = Fight-Flight-Freeze System; Fr = Freeze; MPI = Motor Planning Interruption; BD = Behavioural Disengagement; OT = Obsessive Thought; CRA = Cautious Risk Assessment; RI = Reward Interest; GDP = Goal Drive Persistence; RReact = Reward Reactivity; Imp = Impulsivity; FFA = Fight-Flight-Avoid; J = Joy. Int = Interest; Act = Activation; Ticks and crosses indicate whether the scale is present or absent.

First, Carver and White's (1994) BIS/BAS scales (hereafter, Carver-White Scales) were one of the earliest RST questionnaires to be developed and remain the most widely used (>9500 citations, Google Scholar). It offers a unidimensional measure of BIS, comprising items referring to anxiety in response to potentially punishing events (e.g., *If I think something unpleasant is going to happen, I usually get pretty 'worked up'*). Although not anticipated a-priori, factor analyses suggested a multidimensional BAS, yielding three subscales which the authors termed Drive (e.g., *I go out of my way to get things I want*), Fun Seeking (e.g., *I crave excitement and new sensations*), and Reward Responsiveness (e.g., *When I'm doing well at something I love to keep at it*). Carver and White (1994) found moderate associations ($r_s > .39$) between the three BAS scales and Eysenck et al.'s (1985) Extraversion scale, whereas the BIS scale was associated with multiple measures of trait anxiety ($r_s > .58$). These observations were generally consistent with Gray's proposal as depicted in Figure 1.

Torrubia et al. (2001) subsequently developed the Sensitivity to Punishment and Sensitivity to Reward Questionnaire (SPSRQ; >1500 citations, Google Scholar). The development of the SPSRQ was inspired by several concerns raised by Heubeck et al. (1998) about the Carver-White scales, concerning unexpected positive correlations between BAS and neuroticism, and between BIS and externalising behaviours. The SPSRQ comprised two unidimensional scales labelled Sensitivity to Punishment (e.g., *I worry about things I said or did*) and Sensitivity to Reward (e.g., *I do things for quick gains*). Although these scale labels were reasonable (indeed, labelling a trait scale 'BIS' or 'BAS' arguably conflates traits [e.g., anxiety] with their putative underlying causes [e.g., BIS], see DeYoung, 2015, pp. 36–37), the departure from the earlier 'BIS' and 'BAS' labels perhaps introduced a jangle fallacy into the literature. The SPSRQ was originally developed in Spanish, and English translations yielded persistent comprehension issues, finally motivating a revised and clarified English language version, again comprising two unidimensional scales (Conner et al., 2018).

After Gray and McNaughton's (2000) revision to RST, researchers grew concerned that existing RST questionnaires were based on an out-dated theory. The first questionnaire designed specifically to measure r-RST was the Jackson-5 (Jackson, 2009). Like the SPSRQ, this contained unidimensional scales to assess the BAS and BIS, and like the Carver-White scales, these scales were named for their presumed underlying system. The Jackson-5 also included scales to assess trait manifestations of the FFFS, this being the first attempt to do so since the unsuccessful Gray-Wilson Personality Questionnaire (Wilson et al., 1989; see Table 1). Specifically, three scales were offered to assess distinct dimensions of Fight, Flight,

and Freeze. Whereas Fight described quick, defensive response to proximal threat (e.g., *When provoked, I easily get into a fight*), responses to more distal threats were captured by Flight (e.g., *If a dog barks at me, I would run away*) or Freezing (e.g., *If something bad was about to happen to me, I would just stop*). Over the last decade the Jackson-5 has become a commonly used r-RST questionnaire (>200 citations, Google Scholar).

Motivated by concerns about the validity of existing measures of r-RST, Corr and Cooper (2016) developed the Reinforcement Sensitivity Theory of Personality Questionnaire (RST-PQ). This questionnaire comprises two defensive factors, Flight-Freezing-Avoid (which divides into Flight, Freeze, and Active Avoidance), where Active Avoidance relates to items that reflect phobic avoidance (e.g., *I would not hold a spider or snake*), and BIS. The BIS divides into Motor Planning Interruption (e.g., *I take a long time to make decisions*), Cautious Risk Assessment (e.g., *I worry a lot*), Obsessive Thought (e.g., *My mind is dominated by recurring thoughts*), and Behavioural Disengagement (e.g., *I often feel depressed*), with each factor relating to a specific stage of defensive approach. Meanwhile, BAS comprises four factors: Reward Interest (e.g., *I am always on the go*), Goal-Drive Persistence (in the pursuit of goals, e.g., *I feel driven to succeed in chosen career*), Reward Reactivity (e.g., *I am especially sensitive to reward*), and Impulsivity (lack of restraint, e.g., *I sometimes cannot stop myself talking*). The RST-PQ has rapidly become one of the most prominent r-RST questionnaires (>400 citations, Google Scholar), has been translated into other languages, such as Swedish (Eriksson et al., 2019) and German (Pugnaghi et al., 2018), and was recently complemented by a short-form questionnaire (RST-PQ-S; Vecchione & Corr, 2021).

It is clear from this summary of just four RST questionnaires (fewer than half of those depicted in Table 1), that the field has not converged on an agreed psychometric structure for RST traits, nor a consensus around how to measure them (see Corr, 2001, 2016). Moreover, the accumulation of at least 63 individual trait scales inspired by RST suggests that construct/scale proliferation is an acute problem in this area. Several RST scales appear to provide either divergent conceptualisations of the same trait (jingle) or overlapping conceptualisations of traits labelled differently (jangle; see Table 2 for examples). This is a chaotic state of affairs, particularly for non-experts seeking to measure individual differences relating to RST. Although various reviews have qualitatively compared the various RST questionnaires (e.g., Corr, 2016; Torrubia et al., 2008; Walker & Jackson, 2017), these efforts have not yielded concrete solutions. This situation might be improved if the various RST scales were compared empirically within a common, overarching framework.

Table 2. Examples of jingle-jangle fallacies in RST measurement.

Jingle fallacy		Jangle fallacy	
Jackson-5: BIS Scale	RST-PQ: BIS Scale	Carver-White: Fun Seeking Scale	RST-PQ: Impulsivity Scale
I want to do well compared to my peers I like my peers to know I am doing well	I have spent a lot of time on my own trying to 'get away from it all' I sometimes feel 'blue' for no good reason	I will often do things for no other reason than they might be fun I often act on the spur of the moment	I often do risky things without thinking of the consequences I find myself doing things on the spur of the moment

Note. BIS = Behavioural Inhibition System. RST-PQ = Reinforcement Sensitivity Theory – Personality Questionnaire.

The Big Five as an organising framework

It has long been argued that the Big Five personality taxonomy (comprising extraversion, neuroticism, conscientiousness, agreeableness, and openness) provides a comprehensive organising framework for personality constructs, helping researchers to navigate trait space more easily (e.g. [John et al., 2008](#)). In other words, the Big Five may provide a conceptual space in which personality scales can be located—even those developed entirely outside of the Big Five tradition. The Big Five may thus offer a useful framework to organise RST scales and assess their similarities and differences. Notably, the Big Five includes [Eysenck's \(1967\)](#) 'big two' dimensions (i.e., extraversion and neuroticism) to which [Gray \(1981, 1987\)](#) proposed that trait manifestations of the RST systems should relate (see [Figure 1](#)). One useful feature of the Big Five is its hierarchical structure, with five broad 'domains' positioned above ten 'aspects' ([DeYoung et al., 2007](#)), which are situated in turn above three ([Soto & John, 2017](#)), six ([Costa & McCrae, 1992b](#)), or perhaps many more ([Irwing et al., 2023](#)) narrower 'facets'. Known as the bandwidth-fidelity trade-off ([Gleser et al., 1965](#); [Soto & John, 2017](#)), the broad domains offer broader but less precise personality descriptions (high bandwidth), whilst the narrow facets offer precise but narrow trait description (high fidelity). Thus, locating a scale within the Big Five framework need not require that it is closely related to or redundant with one of the Big Five domains, as it might alternatively be represented as a facet of one more domains. For example, the personality construct and scale known as 'Grit', initially proposed as an ostensibly novel and distinctive trait, is arguably a facet of Big Five conscientiousness (e.g., [Bainbridge et al., 2022](#); [Ponrock et al., 2020](#); [Schmidt et al., 2018](#)).

The usefulness of the Big Five as an organising framework for psychological scales was recently demonstrated by [Bainbridge et al. \(2022\)](#). Like previous authors (e.g., [Paunonen & Jackson, 2000](#); [Saucier & Goldberg, 1998](#)), they used the amount of variance (R^2) that the Big Five domains explained in a given scale as an index of how readily that scale can be incorporated into the Big Five framework. However, whereas previous authors used somewhat arbitrary criteria for deciding what R^2 was sufficient for claiming that a

scale was 'part of' the Big Five (e.g., an R^2 of .20; [Paunonen & Jackson, 2000](#)), [Bainbridge et al. \(2022\)](#) compared the R^2 overlap the Big Five had with a stand-alone scale to the overlap it had with existing facets of the Big Five. They found that around 70% of a set of 40 highly-cited scales were as strongly related to the Big Five as more than 10% (i.e., 3 of 30) of a set of Big Five facets. Among other insights from this work, several scales emerging from different literatures were discovered to converge closely, in terms of their relation to the Big Five dimensions. For example, scales measuring optimism ([Scheier & Carver, 1985](#); [Scheier et al., 1994](#)) and self-esteem ([Rosenberg, 1965](#)) occupied near identical locations within the space of the Big Five (closely aligned with low neuroticism, and secondarily with higher extraversion), despite their distinct conceptual origins. Exploring the locations of various RST scales within the Big Five taxonomy could similarly illuminate the overlap and distinctiveness of these various scales.

Of note, one RST questionnaire (the Carver-White scales) was included among the scales examined in the paper by [Bainbridge et al. \(2022\)](#). Broadly in line with Gray's suggestion, as depicted in [Figure 1](#), the Carver-White BIS scale was strongly aligned with neuroticism whereas two of the BAS scales (Drive and Fun Seeking) were primarily aligned with extraversion. (The third, Reward Responsiveness, was much more interstitial to the Big Five.) In addition, latent variable correlations suggested that two of the Carver-White scales were very highly correlated with existing Big Five facets, perhaps verging on redundancy: the BIS scale was correlated at $r = .90$ with the Anxiety facet of neuroticism, and the BAS Fun Seeking scale was correlated at $r = .94$ with the Excitement Seeking facet of extraversion. We will adopt [Bainbridge et al.'s \(2022\)](#) approach in the current research, which seeks to replicate the findings pertaining to the Carver-White Scales and extend upon these analyses through the inclusion of a more comprehensive set of RST questionnaires.

Summary and aims

The succession of RST questionnaires that have been introduced to the literature provides a striking example of scale proliferation and potential jingle-jangle fallacies.

One way to organise and compare these scales would be to explore their locations within the broad trait space described by the Big Five. To our knowledge, no previous studies have empirically assessed the utility of the Big Five as an organising framework for RST scales. In this paper, we sought to investigate whether RST scales can be located within the space described by the Big Five, potentially as facet-level traits. We adopted a two-stage approach for our investigation, first conducting a largely exploratory study, results of which were used to derive hypotheses that we tested in a second, confirmatory study (for the Supplement, pre-registrations, analysis code, all data for both studies, see <https://osf.io/q9vsk/>).

Study 1

In Study 1 we examined the following three questions: First, how effective is the Big Five as an organising framework for RST scales, as assessed by the variance they share with the Big Five domains, compared to the Big Five facets (RQ1)? Second, where can RST scales be located within the Big Five Personality framework, in terms of the domains with which they are most strongly aligned (RQ2)? Finally, of the RST scales that are locatable within the Big Five Personality framework, are any highly correlated with any existing Big Five facets (RQ3)? Furthermore, endeavouring to replicate findings from Bainbridge et al. (2022), we pre-registered two formal hypotheses: first, that the Carver-White BIS scale will be highly correlated ($r > .60$) with Anxiety, a facet of neuroticism (H1), and second, that the Carver-White BAS Fun Seeking scale will be highly correlated ($r > .60$) with Excitement Seeking, a facet of extraversion (H2).

Method

Participants and procedure. We set a minimum target sample size of $N = 350$, guided by the following considerations: First, we wanted to achieve at least 80% power to detect small-medium effect sizes (i.e., $r = .20 / d = .40$, according to effect-size guidelines for personality research; Gignac & Szodorai, 2016), which would require a minimum of 327 participants for a multiple regression with five predictors (according to G*Power; Faul et al., 2007). Second, we wanted to ensure stable estimates of effect sizes of this magnitude, which would require at least 250 participants (Schönbrodt & Perugini, 2013). Third, we heeded sample size recommendations for exploratory factor analysis and structural equation modelling research, which typically advise a minimum 200–300 observations (e.g., Barrett, 2007; Comrey & Lee, 1992). Finally, we sought to collect as large a sample as possible within in the first University semester of 2023. Ultimately, we recruited 689 participants between April and June, 2023. All

participants were first-year undergraduate psychology students enrolled at a large Australian University who participated online for course credit.

Of our initial sample of 689 participants, two participants were removed as they were under 18 years of age (and thus not covered by our ethical approval). 132 participants appeared to have clicked on the study link but provided few or even no responses and were thus removed. A further 132 participants were then excluded for failing at least one of a series of quality control checks (adapted from Wood et al., 2017). These assessed ‘straight-lining’ (i.e., highly invariant responses to questionnaire items); implausibly fast responses (i.e., < 1 sec per questionnaire item); failing to respond to a simple open-ended question; and (low) self-reported data quality (e.g., “I rushed through this survey”). One participant provided an ambiguous text answer for age and was removed. A further ten cases were discovered to be duplicates and were thus removed. Finally, four participants were excluded for failing to answer five or more items across the survey. Following exclusions (described further in Supplemental section A), the final sample comprised 408 participants (80.6% female), aged 18–52 ($M = 19.22$, $SD = 2.84$), most of whom identified as being of Asian (55.3%) or Caucasian (41.4%) descent. All students from non-English speaking countries had passed an acceptable level of the International English Language Test (<https://www.ielts.org/>), as required for admittance into their university course. Ethical approval was granted by the University’s Office of Research Ethics and Integrity (approval # 2023-25937-38611-5) and all participants provided informed consent.

Materials

Big Five questionnaires. The Big Five Inventory-2 (BFI-2; Soto & John, 2017) was used to measure the Big Five domains. The BFI-2 is a revised and expanded version of the original BFI (John et al., 1991) comprising 12 items for each domain. To measure the Big Five facets we used a 120-item analogue of the NEO Personality Inventory Revised (NEO-PI-R; Costa & McCrae, 2008), drawn from the International Personality Item Pool (i.e., IPIP; <https://ipip.ori.org>) and known as the IPIP-120 (Maples et al., 2014). This questionnaire comprises four items to measure each of the six facets of each Big Five domain (30 facet scales in total). For both questionnaires, participants were asked to rate the extent to which they agreed that various trait descriptions (e.g., *is outgoing*, *sociable*) were applicable to them. Responses were made using a five-point Likert scale ranging from 1 = *Strongly disagree* to 5 = *Strongly agree*. Each questionnaire was presented in a separate block and items for the BIS, BAS, and FFFS scales of each questionnaire were randomised within each block. By intermingling the scales within each block, validity was maximised as each item functioned as an independent assessment of the latent construct (Clifton, 2020).

RST questionnaires. Participants completed the [Carver and White \(1994\)](#) questionnaire, pertinent to our preregistered confirmatory hypotheses. Additional RST questionnaires were then identified based on high citations in Google Scholar—one additional questionnaire based on the original RST, and two questionnaires based on the subsequently revised RST. In total, the four RST questionnaires comprised 27 scales. Responses to items on all questionnaires were made using a similar same Likert scale described for the BFI-2 and IPIP-120 (i.e., 1 = *Completely disagree*; 5 = *Completely agree*), except for the Carver-White scale, which used a 4-point Likert scale (1 = *Very false for me* to 4 = *Very true for me*). Specific details of each questionnaire are shown in [Table 3](#).

Data analyses. Following [Bainbridge et al. \(2022\)](#), we assessed the degree to which RST scales can be located within the space described by the Big Five domains (RQ1) as follows: We first ran a series of latent variable regression

analyses in which each of the IPIP-120 Big Five Facets served as criterion variables and the BFI-2 Big Five domains as simultaneous predictor variables, then ranked the 30 scales from highest to lowest R^2 values. We then ran equivalent latent variable regression analyses with each of the RST scales as criterion variables and BFI-2 Big Five domains as simultaneous predictor variables. We regarded any RST scale as locatable within the space described by the Big Five if its R^2 overlap with the five domains exceeded that of the 27th facet; otherwise, we regarded the scale as “peripheral to the Big Five”. This threshold is obviously stringent, as it implies that three existing facets of the Big Five are somehow not locatable within that framework. Nevertheless, it insures against the possibility that 10% (i.e., 3 of 30) of the facet scales will have an unexpectedly low overlap with the Big Five domains (e.g., due to sampling variability). In addition to this primary threshold, we derived additional thresholds for describing the degree of relatedness between the RST scales and Big Five domains, relative to that of Big Five facets, based on the 10th (labelled as “highly reflective of the Big Five”), 20th (“reflective of the Big Five to an average degree”) and 30th (“least related facet”) facet (see [Bainbridge et al., 2022](#)).

For these analyses, latent variables were preferred to mean-scores given that they represent the shared variance among the items for each factor, thereby adjusting for item-specific measurement error ([Beaujean, 2014](#)). Nevertheless, all analyses were repeated based on mean-scores, for comparative purposes (reported in [Supplemental section B](#)). We used Maximum Likelihood estimation, which is robust to violations of normality ([Widaman, 2012](#)), and oblique rotation, because it is unrealistic to assume that the Big Five domains are uncorrelated ([Bainbridge et al., 2022](#); [DeYoung, 2006](#)). Nevertheless, orthogonal rotation may allow for easier interpretation of RST scale locations within the Big Five, thus equivalent analyses using orthogonal rotation are also reported in [Supplemental section B](#).

To assess *where* the RST scales can be located within the space described by the Big Five domains (RQ2), we inspected the latent regression coefficients (β) within the latent variable models in which the RST scales were regressed on the Big Five domains. To assess whether any RST scales were closely related to any existing Big Five facets (RQ3), we identified any RST scales from the aforementioned models with $\beta \geq .60$, and then computed latent variable correlations to assess the strength of association between those RST scales and the *facets* of the Big Five domain(s) in question. Additionally, to test our preregistered hypotheses (H1 and H2), we examined the latent variable correlations between the Carver-White BIS scale and the Anxiety facet of neuroticism, and between the Carver-White Fun Seeking scale and the Excitement Seeking facet of extraversion.

Table 3. Details of questionnaires used to measure o-RST and r-RST (Study 1).

Theory	Questionnaire	Scales
o-RST	Carver-White Scales Carver and White (1994)	BIS = 7 items Total BAS = 13 items <i>Drive</i> = 4 items <i>Fun Seeking</i> = 4 items <i>Reward Responsiveness</i> = 5 items
o-RST	SPSRQ Conner et al. (2018)	Sensitivity to Punishment (SP) = 10 items Sensitivity to Reward (SR) = 10 items
r-RST	Jackson-5 Scales Jackson (2009)	BIS = 6 items BAS = 6 items Total FFFS = 18 items <i>Fight</i> = 6 items <i>Flight</i> = 6 items <i>Freeze</i> = 6 items
r-RST	RST-PQ Corr and Cooper (2016)	Total BIS = 23 items <i>Motor Planning Interruption (MPI)</i> = 5 items <i>Cautious Risk Assessment (CRA)</i> = 4 items <i>Obsessive Thought (OT)</i> = 8 items <i>Behavioural Disengagement (BD)</i> = 6 items Total BAS = 32 items <i>Reward Interest (RI)</i> = 7 items <i>Goal-Drive Persistence</i> = 7 items <i>Reward Reactivity</i> = 10 items <i>Impulsivity</i> = 8 items Total FFA = 10 items <i>Flight</i> = 3 items <i>Freezing</i> = 5 items <i>Active Avoidance</i> = 2 items

Extending from RQ3, we also explored whether any RST scales approached *redundancy* with existing Big Five facets. Specifically, we applied criteria suggested by [Lawson and Robins \(2021\)](#) for identifying ‘twin constructs’, and selected any RST scale whose latent variable correlation with a Big Five facet exceeded .80. As per [Lawson and Robins \(2021\)](#) and [Irwing et al. \(2023\)](#), we then used confirmatory factor analysis to compare a single-factor model (all RST scale and facet scale items loading on one factor) with a two-factor model (RST scale and facet scale items loading on separate, correlated factors).

All analyses were deployed within *R* (Version 2023.06.2 + 561) using the *Lavaan* package ([Rosseel, 2012](#)), except for data cleaning and computing large intercorrelation matrices (*SPSS* version 26; *JASP* version 0.12.1). Additional *R* packages are reported in [Supplemental section A](#).

Results

Descriptive statistics, [McDonald’s \(1999\)](#) ω reliability estimates and Pearson and Spearman correlations between the Big Five domains (as measured using the BFI-2) and (a) facet scales (as measured using the IPIP-120), and (b) RST scales, are reported in [Supplemental section B](#). Reliability estimates for the IPIP-120 scales were generally acceptable ($\omega > .70$) and each IPIP-120 facet had positive correlations exceeding .20 with their respective BFI-2 domain.² Of note, ω was below .70 for a handful of BAS and FFFS scales, indicating low reliability.

RQ1: How effective is the Big Five as an organising framework for RST scales? Just over 70% of the RST scales surpassed the 27th facet, indicating that they could plausibly be located within the Big Five. The average R^2 of the RST scales predicted by the Big Five domains was .33, compared with .47 for the facets. [Table 4](#) shows the percentage of RST scales that passed each criterion. An equivalent table showing the percentage of RST scales that passed the R^2 criteria based on analyses of mean-scores can be found in [Supplemental section B](#). Results based on mean-scored

scales were largely consistent with the results from the latent variable regressions.

The R^2 value for each RST scale from the latent variable regressions is depicted in [Figure 2](#) (horizontal bars) against each of the criteria noted in [Table 4](#) (vertical dotted lines). Numerical R^2 values used to generate this figure can be found in [Supplemental section B](#). This firstly reveals that most BIS scales (88%) were locatable within the Big Five and were deemed to be at least as related to the Big Five as the 20th facet (‘average’). Indeed, the five RST scales that were as related to the Big Five as the 10th facet (‘highly reflective’) were all BIS scales. Of these, the RST-PQ Behavioural Disengagement scale was the most highly reflective of the Big Five. Conversely, the Jackson-5 BIS scale was the only BIS scale with an R^2 value below that of the 27th facet (‘peripheral’), indicating that it might not be accommodated within the space of the Big Five dimensions at all. Second, nine BAS scales (82%) were at least as related to the Big Five as the 27th facet, and five of these exceeded the 20th facet (‘average’). Only two BAS scales—Carver-White Reward Responsiveness and RST-PQ Reward Reactivity—were unable to be located within the Big Five (‘peripheral’). Finally, only three FFFS scales (38%) were locatable within the space of the Big Five. Of these, Jackson-5 Freeze was as related to the Big Five as the 20th facet (‘average’), whilst the other two scales (Jackson-5 Fight, and the total Jackson-5 FFFS), were deemed to be ‘somewhat independent’. Remaining FFFS scales from both the Jackson-5 and the RST-PQ did not appear to be locatable within the Big Five (‘peripheral’).

RQ2: Where are RST scales located within the Big Five? [Figure 3](#) depicts regression coefficients from the latent variable models described above. These describe the unique relation each Big Five domain has with each RST scale, and thus the location of each RST scale within the space described by the Big Five. Note that scales are ordered from the largest R^2 (RST-PQ Behavioural Disengagement) to the smallest R^2 (RST-PQ Active Avoidance), and we focus only on scales deemed to fall within the Big Five (i.e., exceeding the 27th facet), as indicated by the dotted line. (An equivalent Figure based on mean-scores can be found in [Supplemental section B](#)). First, all BIS scales exceeding the 27th facet had primary positive associations with neuroticism ($\beta > .50$). Second, most of the BAS scales deemed locatable within the Big Five were moderately to strongly associated with extraversion ($\beta > .30$), and one shared a primary, positive association with conscientiousness ($\beta > .50$). Finally, the three FFFS scales deemed locatable within the Big Five—all from the Jackson 5 Scales—had more complex patterns of association.

Table 4. Cumulative percentage of RST scales that passed criterion (Study 1).

Facet criteria	Category	R^2	% of RST scales above criterion
10th facet	Highly reflective	.57	18.52% (5)
20th facet	Average	.36	44.44% (12)
27th facet	Somewhat independent	.24	70.37% (19)
30th facet	The least related facet	.09	92.59% (25)

Note. Criteria compared against R^2 criterion derived as per [Bainbridge et al., \(2022\)](#). The number of RST scales that passed each criterion is indicated in parentheses. Scales were assessed against the criteria before rounding to two decimal places.

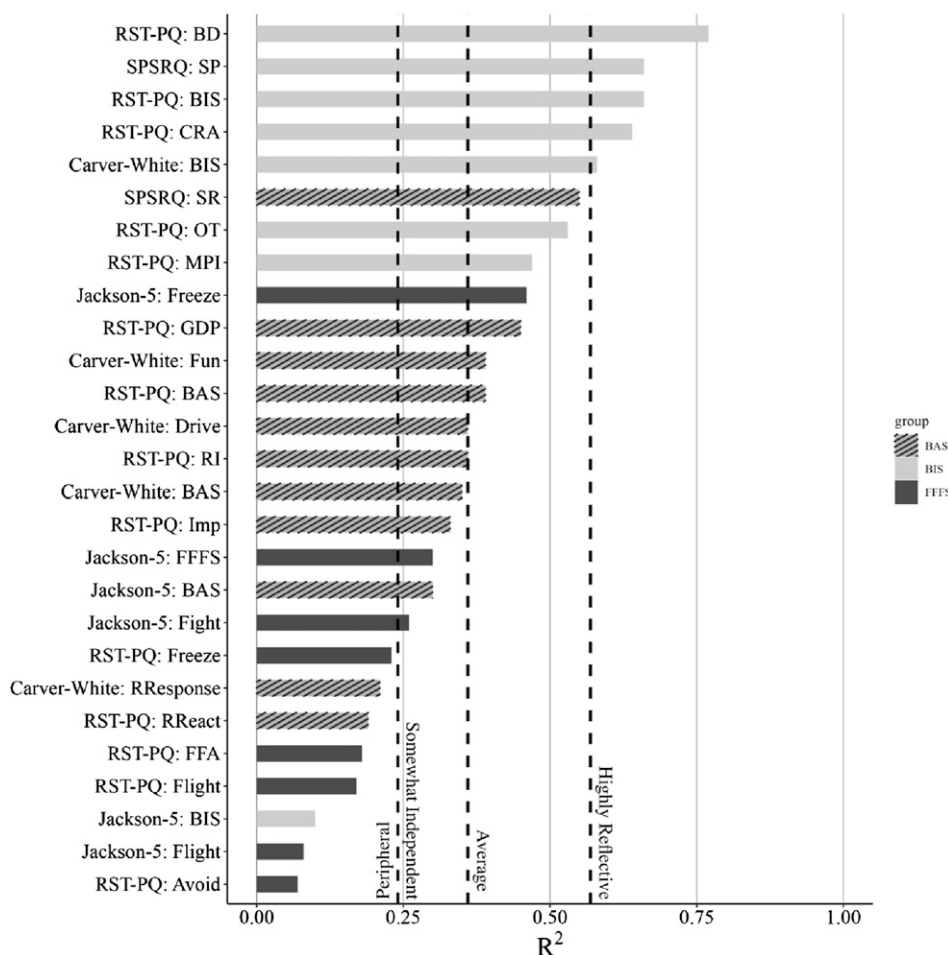


Figure 2. Latent variable regression R^2 values for RST scales (Study I). *Note.* RST-PQ = Reinforcement Sensitivity Theory – Personality Questionnaire. BD = Behavioural Disengagement. BIS = Behavioural Inhibition System. SPSRQ = Sensitivity to Punishment, Sensitivity to Reward Questionnaire. SP = Sensitivity to Punishment. CRA = Cautious Risk Assessment. SR = Sensitivity to Reward. OT = Obsessive Thought. MPI = Motor Planning Interruption. FFFS = Fight-Flight-Freeze System. GDP = Goal-Drive Persistence. BAS = Behavioural Activation System. Fun = Fun Seeking. RI = Reward Interest. FFA = Flight-Freeze-Avoid. RResponse = Reward Responsiveness. RReact = Reward Reactivity. Avoid = Active Avoidance. See [Supplemental section B](#) for the R^2 values extracted to make this figure.

Specifically, both the Fight and total FFFS scales had primary, negative associations with agreeableness, and positive associations with extraversion. Conversely, the Freeze scale had strong positive associations with neuroticism, and a weaker, negative association with extraversion.

RQ3: Are any RST scales closely associated with existing Big Five Facets? In support of both of our pre-registered hypotheses, [Carver and White's \(1994\)](#) BIS and Fun Seeking scales had large latent variable correlations ($r > .60$) with the Anxiety facet of neuroticism and the Excitement Seeking facet of extraversion, respectively (see [Table 5](#)). Indeed, these correlations both approached .80, suggesting they might be considered ‘twin constructs’ ([Lawson & Robins, 2021](#)). Equivalent correlations based on mean-scored scales showed a very similar pattern (see [Supplemental section B](#)).

Turning to our more exploratory analyses, [Table 6](#) presents all RST scales that had both strong, unique associations with at least one BFI-2 domain ($\beta \geq .60$) and a latent variable correlation ($r \geq .60$) with at least one IPIP-120 facet. Almost all of these were BIS scales, and were all highly correlated with three facets of neuroticism—Anxiety, Depression, and Self-Consciousness. No BAS nor FFFS scales appear in [Table 6](#), which accords with our finding that these scales were generally less reflected in the Big Five.

Are any RST scales redundant with existing Big Five facets? As is clear from [Table 6](#), four RST scales appeared to be approaching redundancy with existing Big Five facets, as indicated by latent correlations of $r \geq .80$. We therefore ran a series of confirmatory factor analyses to compare the fit of single-factor models to two-factor correlated models for

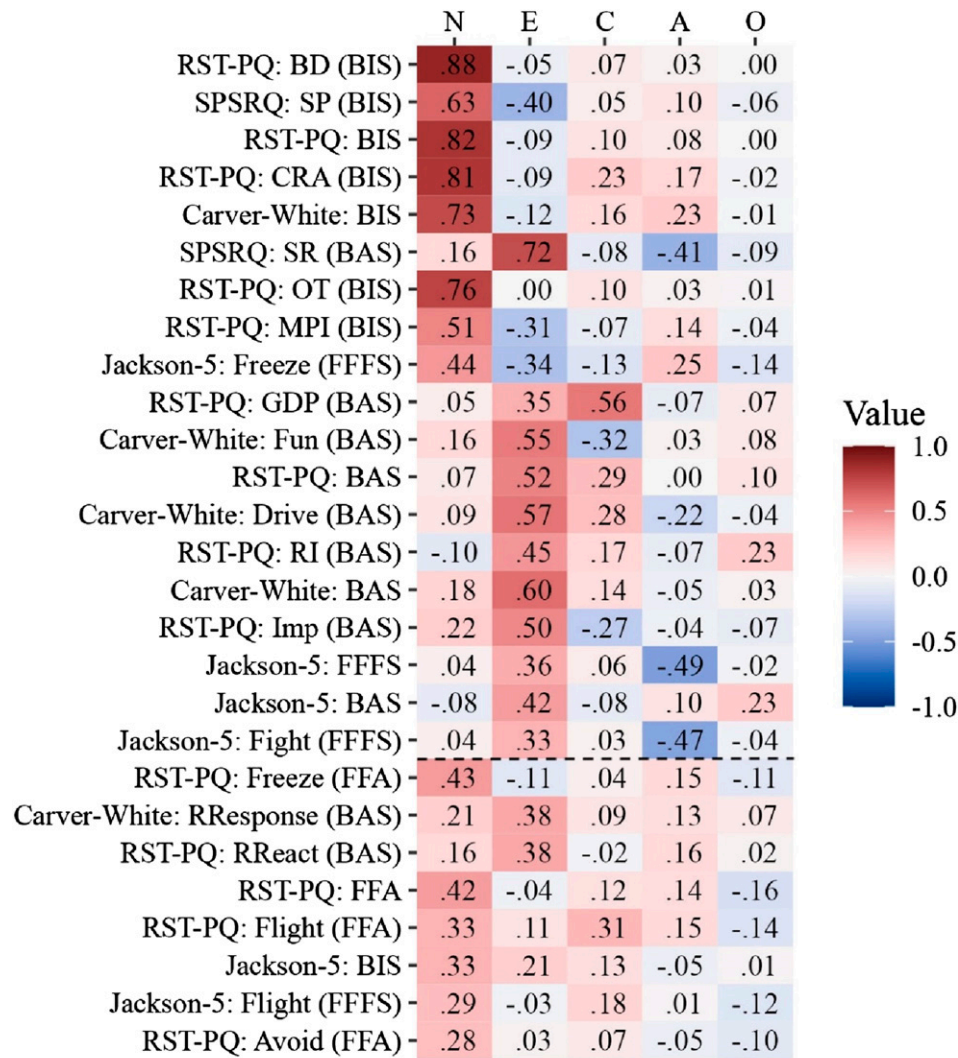


Figure 3. RST scales' standardised locations within the Big Five (Study 1). *Note.* Scales below the dashed line are deemed to be peripheral to the Big Five (see Figure 2). E = extraversion. A = agreeableness. N = neuroticism. O = openness. C = conscientiousness. RST-PQ = Reinforcement Sensitivity Theory – Personality Questionnaire. BD = Behavioural Disengagement. BIS = Behavioural Inhibition System. SPSRQ = Sensitivity to Punishment, Sensitivity to Reward Questionnaire. SP = Sensitivity to Punishment. CRA = Cautious Risk Assessment. SR = Sensitivity to Reward. OT = Obsessive Thought. MPI = Motor Planning Interruption. FFFS = Fight-Flight-Freeze System. GDP = Goal-Drive Persistence. BAS = Behavioural Activation System. Fun = Fun Seeking. RI = Reward Interest. Imp = Impulsivity. FFA = Flight-Freeze-Avoid. RResponse = Reward Responsiveness. RReact = Reward Reactivity. Avoid = Active Avoidance.

Table 5. Preregistered correlations between RST scales and Big Five facets (Study 1).

Carver-White scale	IPIP-120 facet	Latent variable correlation	90% CI [LL, UL]
BIS	N: Anxiety	.76	[.71, .81]
Fun Seeking (BAS)	E: Excitement Seeking	.79	[.72, .87]

Note. IPIP-120 = International Personality Item Pool. BIS = Behavioural Inhibition System. N = neuroticism. E = extraversion. CI = Confidence Interval. LL = Lower Limit. UL = Upper Limit.

each of these pairs of scales (see Irwing et al., 2023; Lawson & Robins, 2021). As shown in Table 7, all of the two-factor solutions showed better fit than the single-factor models, although fit was often unacceptable for both models. For instance, both the single-factor and two-factor models for the RST-PQ Cautious Risk Assessment and Anxiety scales have acceptable comparative fit ($>.95$; Hu & Bentler, 1999), and absolute fit ($<.08$; Hu & Bentler, 1999), whilst the parsimony-adjusted indices suggest that the single-factor model has unacceptable fit. For each pair of models, the difference between their chi-

Table 6. RST scales with strong correlations ($r \geq .60$) with existing Big Five Facets (Study 1).

RST scale	Big Five facet	Latent variable correlation	90% CI [LL, UL]
RST-PQ: CRA (BIS)	N: Anxiety	.88	[.85, .91]
SPSRQ: SP (BIS)	N: Self-Consciousness	.85	[.80, .90]
RST-PQ: BD (BIS)	N: Depression	.82	[.78, .86]
RST-PQ: BIS	N: Anxiety	.82	[.78, .85]
RST-PQ: OT (BIS)	N: Anxiety	.76	[.71, .81]
SPSRQ: SP (BIS)	N: Anxiety	.73	[.68, .78]
Carver-White: BIS	N: Self-Consciousness	.72	[.65, .79]
RST-PQ: BD (BIS)	N: Anxiety	.70	[.64, .75]
RST-PQ: BIS	N: Depression	.66	[.61, .72]
SPSRQ: SP (BIS)	N: Depression	.62	[.56, .68]
RST-PQ: CRA (BIS)	N: Self-Consciousness	.62	[.54, .59]
RST-PQ: BIS	N: Self-Consciousness	.61	[.54, .68]
RST-PQ: OT (BIS)	N: Depression	.61	[.55, .67]

Note. IPIP-120 = International Personality Item Pool. N = neuroticism. BIS = Behavioural Inhibition System. CRA = Cautious Risk Assessment. SP = Sensitivity to Punishment. BD = Behavioural Disengagement. OT = Obsessive Thought. BAS = Behavioural Activation System. SR = Sensitivity to Reward. CI = Confidence Interval. LL = Lower Limit. UL = Upper Limit.

square values, which can be used to assess whether there is a significant difference in fit between the two-factor and single-factor models, exceeded the critical value of $\chi^2 = 3.84$. This indicates that the two-factor models did offer significantly improved fit. However, the improvement in model fit (which comes at a cost to parsimony) was generally modest.

Summary. Study 1 revealed that a majority of RST scales examined overlapped with the Big Five to a degree suggesting these scales might plausibly be represented as Big Five facets (RQ1). Whereas BIS scales were principally aligned with neuroticism, BAS scales were generally aligned with extraversion but also with conscientiousness (RQ2). Several BIS scales were strongly correlated with facets of neuroticism, and some appeared to approach redundancy with these facets (RQ3). Finally, we confirmed previously observed strong associations between the Carver-White BIS scale and the Anxiety facet of neuroticism (H1), and the Carver-White BAS Fun

Seeking scale and the Excitement Seeking, facet of extraversion (H2).

Study 2

In Study 2, we sought to confirm key findings from Study 1 via a pre-registered direct replication. To enhance the generalisability of our Study 1 findings beyond a student sample, we recruited a community sample of adults living in Australia. We also expanded our evaluation of RST scales by adding the Reuter-Montag rRST-Q to our set of questionnaires. Based on our Study 1 findings, we predicted that at least two-thirds (~67%) of included RST scales would be locatable within the space described by the Big Five (H1); that two of the Carver-White BIS/BAS (BIS and BAS Fun Seeking) scales would have very strong latent correlations (i.e., $r \geq .80$) with two existing IPIP-120 facets (Anxiety and Excitement Seeking, respectively), perhaps suggesting redundancy with those facets (H2)³; and that the following pairs of scales would also have very strong latent variable correlations (i.e., $r \geq .80$), suggestive of redundancy (H3): RST-PQ Cautious Risk Assessment and Anxiety, SPSRQ Sensitivity to Punishment and Self-Consciousness, RST-PQ Behavioural Disengagement and Depression, and RST-PQ BIS and Anxiety. The Supplement, preregistration, analysis code, and data can all be found on the OSF (see <https://osf.io/q9vsk/>).

Method

Participants and procedure. As is detailed in our preregistration, we set a target minimum sample size of $N = 400$ for this study and oversampled to ensure a minimum N of 400. We recruited 456 participants via the online crowd sourcing platform, *Prolific*TM in January 2024. All participants were at least 18 years old, located in Australia, spoke English as their primary language, who and participated online for financial reimbursement. Of our initial sample, one participant was excluded for failing to complete at least 25% of Big Five domain items, whilst a further 37 were excluded for failing at least one of a series of quality control checks. This resulted in a total of 38 participant exclusions (9.1% of total sample). More specific details of these exclusions and further demographic information of the final sample are summarised in [Supplemental section A](#). After exclusions the final sample comprised 418 participants, (54.3% female), aged 18–86 ($M = 34.68$, $SD = 11.70$), most of whom identified as being Caucasian (70.6%) or Asian (23.9%). Ethical approval was granted by the University's Office of Research Ethics and Integrity (Ethics ID: 2023-25,937-38,611–5) and all participants provided informed consent.

Materials. Measures of the Big Five domains and facets, and RST scales, were as described in Study 1. As per our pre-

Table 7. Comparison of model fit for single- and two-factor models (Study 1).

Correlated constructs		Single-factor model							Two-factor model					
	IPIP-120 facet scale				RMSEA 90% CI [LL, UL]							RMSEA 90% CI [LL, UL]		
RST scale		χ^2	CFI	TLI		SRMR	BIC	χ^2	CFI	TLI		SRMR	BIC	
RST-PQ: CRA (BIS)	N: Anxiety	126.56	.94	.91	.11 [.10, .13]	.05	7662.88	66.67	.97	.96	.08 [.06, .10]	.03	7609.00	
SPSRQ: SP	N: Self-Consciousness	286.54	.89	.87	.08 [.07, .09]	.05	15,664.92	258.77	.90	.88	.08 [.07, .09]	.05	15,643.16	
RST-PQ: BD (BIS)	N: Depression	435.21	.83	.78	.17 [.15, .18]	.08	10,119.04	311.63	.88	.84	.14 [.13, .16]	.06	10,001.47	
RST-PQ: BIS	N: Anxiety	1504.11	.79	.77	.09 [.09, .10]	.07	25,395.84	1308.62	.82	.81	.09 [.08, .09]	.06	25,206.36	

Note. RST = Reinforcement Sensitivity Theory. IPIP-120 = 120-item International Personality Item Pool. RST-PQ = Reinforcement Sensitivity Theory – Personality Questionnaire. CRA = Cautious Risk Assessment. BIS = Behavioural Inhibition System. N = neuroticism. SPSRQ = Sensitivity to Punishment, Sensitivity to Reward Questionnaire. SP = Sensitivity to Punishment. BD = Behavioural Disengagement. CFI = Comparative Fit Index. TLI = Tucker Lewis Index. RMSEA = Root Mean Square Error of Approximation. CI = Confidence Interval. LL = Lower Limit. UL = Upper Limit. SRMR = Standardized Root Mean Square Residual.

registration, we also added the Reuter-Montag rRST-Q (Reuter et al., 2015), which consists of an 11-item *BIS* scale, 8-item *BAS* scale, 5-item *Fight* scale, 3-item *Flight* scale, and 4-item *Freeze* scale. A total FFFS score, averaging the *Fight*, *Flight*, and *Freeze* items is also computed.

Data analyses. Given that Study 2 is a direct replication of Study 1, we employed the same analyses described earlier (see Study 1 Method). These were again deployed within *R* (Version 2023.06.2 + 561) using the *Lavaan* package (Rosseel, 2012), except for data cleaning and computing large intercorrelation tables (*SPSS* version 26; *JASP* version 0.12.1). Additional packages are reported in Supplemental section A.

Results

Descriptive statistics, reliability estimates, and Pearson correlations between the BFI-2 domains and both the IPIP-120 facets and RST scales are presented in Supplemental section C. The patterns of association were virtually identical to those reported in Study 1. All IPIP-120 facets had positive correlations exceeding .20 with their respective BFI-2 domain, except for the Modesty facet and the BFI-2 Agreeableness domain (though this correlation was similarly low in Study 1).

RQ1: How effective is the Big Five as an organising framework for RST scales? For Hypothesis 1, we predicted that at least two-thirds of RST scales would be locatable within the Big Five, as determined by the criteria applied in Study 1. This was supported when comparing the overlap between the RST scales and the Big Five, relative to the criterion for relatedness of the 27th facet. As per Table 8, somewhat

more of the scales surpassed the 27th facet in Study 2 (79%) than in Study 1 (70%). This was reflected in a higher average R^2 value for the RST scales predicted by the Big Five domains at .44, compared with .37 in Study 1, and .51 for the facets, compared to .47 in Study 1.

Figure 4 depicts the R^2 value for each RST scale from the latent variable regressions as horizontal bars, with each of the criteria indicated with vertical dotted lines. Echoing Study 1, most BIS scales (89%) were locatable within the Big Five, the sole exception again being the Jackson-5 BIS scale, which had an R^2 value below that of the 27th facet ('peripheral'). Of the nine RST scales considered to be 'highly reflective' of the Big Five, two-thirds were BIS scales. Second, ten (83%) BAS scales were at least as related to the Big Five as the 27th facet. Whilst just one BAS scale, from the Reuter-Montag rRST-Q, was considered 'highly reflective' of the Big Five, most of the remaining BAS scales had R^2 values between the 20th and 27th facets. Finally, a majority (67%) of FFFS scales were locatable within the Big Five, in contrast to the sizable minority (38%) observed in Study 1. Two FFFS scales, the Reuter-Montag Flight and Jackson-5 Freeze scales, were considered 'highly reflective' of the Big Five. Nevertheless, the FFFS scales again accounted for the majority of scales deemed 'peripheral' to the Big Five. Interestingly, whilst one Flight scale (the Reuter-Montag Flight scale) was at least as related to the Big Five as the 10th facet, the other two Flight scales (from the Jackson-5 and RST-PQ) were peripheral to the Big Five domains.

RQ2: Where are RST scales located within the Big Five? Figure 5 presents the latent regression coefficients, indicating the RST scales' locations within the Big Five, which are ordered from largest R^2 (RST-PQ Behavioural

Disengagement) to smallest R^2 (RST-PQ Flight). We comment only on scales above the dashed line, which are at least as related to the Big Five as the 27th facet.

As in Study 1, most BIS scales deemed to fall within the Big Five had strong, positive associations with neuroticism ($\beta > .60$), and weak associations across the other four

Table 8. Percentage of RST scales that passed criterion (Study 2).

Facet criteria	Category	R^2	% of RST scales above criterion
10th facet	Highly reflective	.62	21.21% (7)
20th facet	Average	.40	54.55% (18)
27th facet	Somewhat independent	.25	78.79% (26)
30th facet	The least related facet	.17	93.94% (31)

domains. That said, the SPSRQ Sensitivity to Punishment, RST-PQ Motor Planning Interruption and Reuter-Montag BIS scales all had moderate associations across multiple domains, suggesting that they may be interstitial facets. Also mirroring Study 1, all BAS scales surpassing the 27th facet had strong, primary positive associations with extraversion, except for the RST-PQ Goal-Drive Persistence scale, which had a stronger, positive relationship with conscientiousness. Of note, the Reuter-Montag BAS scale had the strongest relationship with extraversion and was minimally associated with the other four domains. Other BAS scales, such as the Carver-White Fun-Seeking and RST-PQ Impulsivity scales, had secondary, negative associations with conscientiousness, whilst the RST-PQ BAS scale had a secondary, positive association. Finally, a diffuse pattern of associations emerged for the Fight, Flight, and Freeze scales: Whereas the Jackson-5 Fight scale did not surpass the 27th facet (contrary to Study

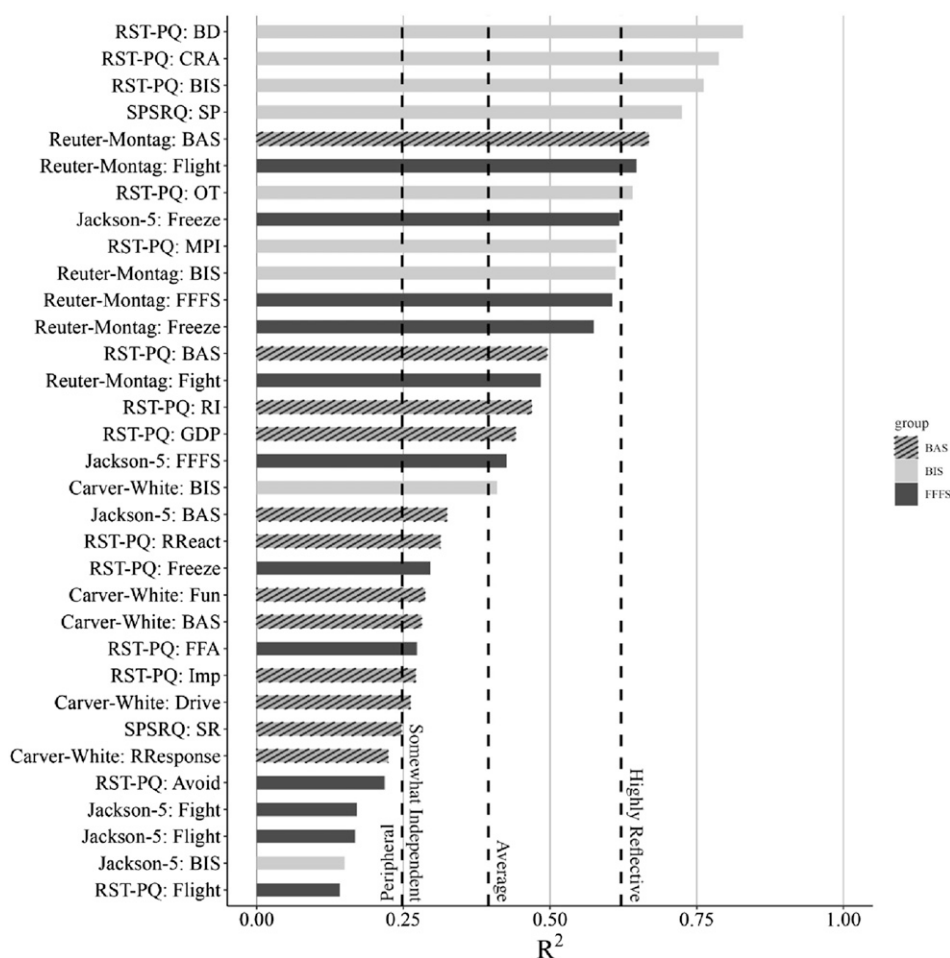


Figure 4. Latent variable regression R^2 values for RST scales (Study 2). *Note.* RST-PQ = Reinforcement Sensitivity Theory – Personality Questionnaire. BD = Behavioural Disengagement. BIS = Behavioural Inhibition System. SPSRQ = Sensitivity to Punishment, Sensitivity to Reward Questionnaire. SP = Sensitivity to Punishment. CRA = Cautious Risk Assessment. SR = Sensitivity to Reward. OT = Obsessive Thought. MPI = Motor Planning Interruption. FFFS = Fight-Flight-Freeze System. GDP = Goal-Drive Persistence. BAS = Behavioural Activation System. Fun = Fun Seeking. RI = Reward Interest. FFA = Flight-Freeze-Avoid. RResponse = Reward Responsiveness. RReact = Reward Reactivity. Avoid = Active Avoidance. See [Supplemental section C](#) for the R^2 values extracted to make this figure.

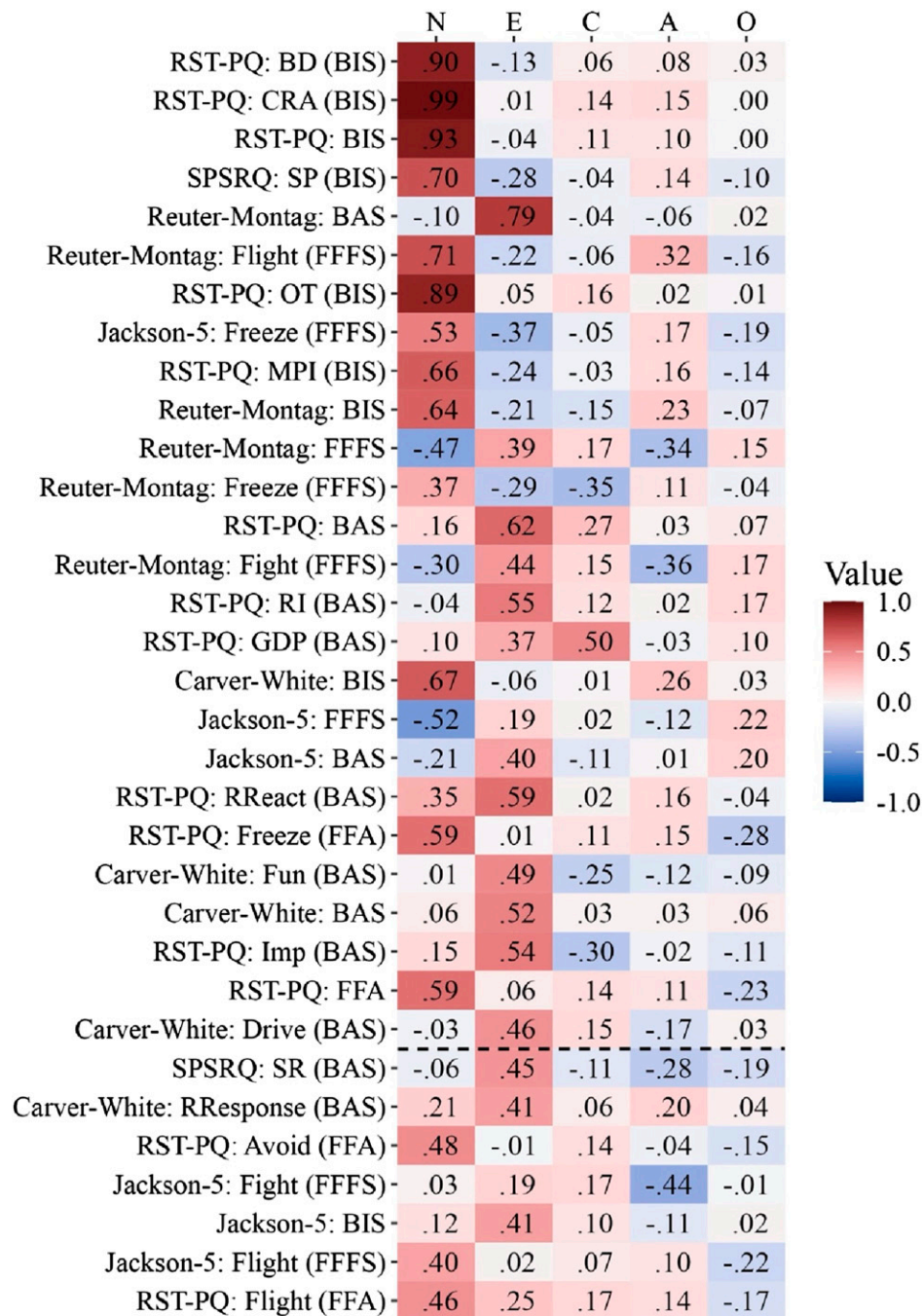


Figure 5. RST scales' standardised locations in the Big Five (Study 2). *Note.* E = extraversion. A = agreeableness. N = neuroticism. O = openness. C = conscientiousness. RST-PQ = Reinforcement Sensitivity Theory – Personality Questionnaire. BD = Behavioural Disengagement. BIS = Behavioural Inhibition System. SPSRQ = Sensitivity to Punishment, Sensitivity to Reward Questionnaire. SP = Sensitivity to Punishment. CRA = Cautious Risk Assessment. SR = Sensitivity to Reward. OT = Obsessive Thought. MPI = Motor Planning Interruption. FFFS = Fight-Flight-Freeze System. GDP = Goal-Drive Persistence. BAS = Behavioural Activation System. Fun = Fun Seeking. RI = Reward Interest. Imp = Impulsivity. FFA = Flight-Freeze-Avoid. RResponse = Reward Responsiveness. RReact = Reward Reactivity. Avoid = Active Avoidance.

1), the Reuter-Montag Fight scale did, and had a strong, positive relationship with extraversion, and reasonably strong associations with the other four domains, suggesting it may be an interstitial facet. Secondly, the only Flight scale to surpass the 27th facet was again from the Reuter-Montag rRST-Q, yielding a strong primary association with neuroticism. Finally, all three Freeze scales were deemed locatable within the Big Five, with the Jackson-5 and Reuter-Montag Freeze scales demonstrating similar primary, positive associations with neuroticism, and secondary, negative relationships with extraversion. Conversely, the RST-PQ Freeze scale had the strongest, positive relationship with neuroticism, but was not associated with extraversion.

RQ3: Are any RST scales closely associated with existing Big Five Facets? For Hypothesis 2, we predicted that the latent variable correlations between the Carver-White BIS scale and Anxiety (a facet of neuroticism), and the Carver-White Fun Seeking scale and Excitement Seeking (a facet of extraversion) would exceed $r = .80$. Both correlations were strong, but not so strong as we had hypothesised (see Table 9).

To test Hypothesis 3, we ran latent variable correlations between the RST scales and IPIP-120 neuroticism facets shown in boldface in Table 10. These correlations confirmed our pre-registered hypotheses that these four latent correlations would be very strong ($r > .80$), thereby replicating our Study 1 findings. Table 10 also shows strong ($r \geq .60$) latent variable correlations between RST scales and IPIP-120 facets. In all, 86% of the strong latent correlations observed in Study 1 were replicated in Study 2.

As in Study 1, Table 10 overwhelmingly comprises strong correlations between BIS scales and neuroticism facets. All BIS scales had high latent correlations with the Anxiety facet of neuroticism, and at least one additional neuroticism facet. Further, all BIS scales, except for the RST-PQ Obsessive Thought scale, had a strong, positive latent correlation with the Self-Consciousness facet of neuroticism. Two BAS scales (the Reuter-Montag BAS, and the RST-PQ BAS scale) were highly correlated with

the Excitement Seeking facet of extraversion. Finally, there were high latent variable correlations between FFFS scales and neuroticism facets. Indeed, the Reuter-Montag Flight scale was very highly correlated ($r > .80$) with Vulnerability. This was somewhat unexpected given the absence of high latent variable correlations involving the BAS and FFFS scales in Study 1.

Finally, we compared the fit of single- to two-factor confirmatory factor analysis models for all scale pairs in Table 10 with latent correlations equal to or exceeding .80 (see Table 11). Of note, the fit of the single-factor for the

Table 10. RST scales with strong correlations ($r \geq .60$) with existing Big Five facets (Study 2).

RST scale	Big Five facet	Latent variable correlation	95% CI [LL, UL]
RST-PQ: BD (BIS)	N: Depression	.93	[.90, .95]
RST-PQ: CRA (BIS)	N: Anxiety	.90	[.87, .94]
RST-PQ: BIS	N: Anxiety	.85	[.81, .88]
SPSRQ: SP (BIS)	N: Self-Consciousness	.84	[.80, .89]
Reuter-Montag: BAS	E: Excitement Seeking	.83	[.78, .88]
Reuter-Montag: Flight (FFFS)	N: Vulnerability	.82	[.74, .90]
SPSRQ: SP (BIS)	N: Anxiety	.80	[.75, .84]
Reuter-Montag: Flight (FFFS)	N: Self-Consciousness	.79	[.70, .87]
RST-PQ: OT (BIS)	N: Anxiety	.78	[.73, .83]
RST-PQ: BIS	N: Depression	.76	[.71, .80]
RST-PQ: MPI (BIS)	N: Self-Consciousness	.75	[.69, .82]
RST-PQ: BD (BIS)	N: Anxiety	.74	[.68, .80]
Reuter-Montag: Flight (FFFS)	N: Anxiety	.74	[.65, .82]
RST-PQ: MPI (BIS)	N: Anxiety	.73	[.67, .80]
Reuter-Montag: BIS	N: Self-Consciousness	.72	[.66, .79]
Reuter-Montag: BIS	N: Anxiety	.72	[.66, .77]
SPSRQ: SP (BIS)	N: Depression	.67	[.61, .73]
RST-PQ: CRA (BIS)	N: Depression	.67	[.60, .73]
RST-PQ: OT (BIS)	N: Depression	.66	[.60, .72]
RST-PQ: BAS	E: Excitement Seeking	.65	[.58, .72]
SPSRQ: SP (BIS)	N: Vulnerability	.65	[.58, .71]
RST-PQ: MPI (BIS)	N: Vulnerability	.64	[.56, .71]
RST-PQ: CRA (BIS)	N: Self-Consciousness	.63	[.55, .71]
RST-PQ: BIS	N: Self-Consciousness	.63	[.56, .70]
Reuter-Montag: BIS	N: Vulnerability	.60	[.53, .67]

Note. IPIP-120 = International Personality Item Pool. N = neuroticism. E = extraversion. BIS = Behavioural Inhibition System. CRA = Cautious Risk Assessment. SP = Sensitivity to Punishment. BD = Behavioural Disengagement. OT = Obsessive Thought. BAS = Behavioural Activation System. MPI = Motor Planning Interruption. CI = Confidence Interval. LL = Lower Limit. UL = Upper Limit.

Table 9. Preregistered latent correlations between RST scales and Big Five facets (Study 2).

Carver-White scale	IPIP-120 facet	Latent variable correlation	95% CI [LL, UL]
BIS	N: Anxiety	.68	[.61, .74]
Fun Seeking (BAS)	E: Excitement Seeking	.78	[.72, .84]

Note. IPIP-120 = International Personality Item Pool. BIS = Behavioural Inhibition System. N = neuroticism. E = extraversion. CI = Confidence Interval. LL = Lower Limit. UL = Upper Limit.

Table 11. Comparison of model fit for single- and two-factor models (Study 2).

Correlated constructs		Single-factor model							Two-factor model						
RST scale	IPIP-120 facet scale	χ^2	CFI	TLI	RMSEA	90% CI [LL, UL]	SRMR	BIC	χ^2	CFI	TLI	RMSEA	90% CI [LL, UL]	SRMR	BIC
RST-PQ: BD (BIS) RST-PQ: CRA (BIS) RST-PQ: BIS SPSRQ: SP (BIS)	N: Depression	524.99	.84	.79	.18	[.17, .20]	.08	8266.49	467.72	.86	.81	.18	[.16, .19]	x	7950.70
	N: Anxiety	161.55	.93	.91	.13	[.11, .15]	.05	16,079.44	103.42	.96	.94	.10	[.08, .10]	.04	16,005.03
	N: Anxiety	1568.04	.82	.80	.10	[.09, .10]	.06	10,119.04	1335.81	.85	.84	.09	[.08, .09]	.06	10,153.71
	N: Self-Consciousness	379.14	.90	.89	.10	[.09, .11]	.05	26,422.28	298.70	.93	.92	.08	[.07, .09]	.05	26,196.09
Reuter-Montag: BAS	E: Excitement Seeking	344.74	.85	.82	.11	[.10, .13]	.07	6940.56	254.21	.90	.87	.10	[.08, .11]	.06	6924.27
Reuter-Montag: Flight (FFFS)	N: Vulnerability	100.78	.93	.89	.12	[.10, .15]	.06	15,904.73	78.45	.95	.91	.11	[.09, .13]	.05	15,653.52
SPSRQ: SP (BIS)	N: Anxiety	654.49	.85	.82	.13	[.13, .14]	.06	10,922.39	397.23	.91	.90	.10	[.09, .11]	.05	10,837.90

Note. RST = Reinforcement Sensitivity Theory. IPIP-120 = 120-item International Personality Item Pool. RST-PQ = Reinforcement Sensitivity Theory – Personality Questionnaire. BD = Behavioural Disengagement. CRA = Cautious Risk Assessment. BIS = Behavioural Inhibition System. N = neuroticism. E = extraversion. SPSRQ = Sensitivity to Punishment, Sensitivity to Reward Questionnaire. SP = Sensitivity to Punishment. BAS = Behavioural Activation System. FFFS = Fight-Flight-Freeze System. CFI = Comparative Fit Index. TLI = Tucker Lewis Index. RMSEA = Root Mean Square Error of Approximation. CI = Confidence Interval. LL = Lower Limit. UL = Upper Limit. SRMR = Standardized Root Mean Square Residual.

RST-PQ BIS and Anxiety facet was better than for the two-factor model based on the BIC. For all models, although the difference between the chi-square values of the single-factor and two-factor models (indicating whether the difference in fit between the models is significant) suggested that the two-factor models again offered significantly improved fit, differences were small and came at a cost to parsimony. Overall, these analyses indicate that several RST scales are approaching redundancy with existing Big Five facets.

Summary. Study 2 largely confirmed our pre-registered hypotheses, garnering further support for the utility of the Big Five as an organising framework for RST scales. At least two-thirds of RST scales were locatable within the Big Five (H1), and four BIS scales were very strongly correlated with facets of neuroticism (H3). We did not find evidence for a very strong latent-correlation between the Carver-White BIS and Fun Seeking scales, and the Anxiety and Excitement Seeking facets, respectively (H2), although these associations were nevertheless strong and similar to that observed in Study 1. Our results suggest that these RST scales (and several others) are virtually redundant with existing facets of the Big Five.

Discussion

Over the past 40 years, the RST literature has become increasingly obscured by a steady proliferation of psychometric scales. Among the dozens of such scales (see Table 1), jingle-jangle fallacies (see Table 2) make it difficult to know how to compare and choose between them. Given that the Big Five has been shown to be a useful organising framework for various trait scales (Bainbridge et al., 2022), we examined whether it might likewise help to organise and synthesise RST scales. Findings from Study 1 were closely replicated in Study 2, and have the potential to bring much-needed clarification to RST measurement.

RQ1: How useful is the Big Five as an organising framework for RST scales?

To evaluate RQ1, we adopted Bainbridge et al.'s (2022) methodology for deriving criteria based on the variance shared between the Big Five domains and facets, reasoning that any RST scale that was as closely related to the Big Five as the 27th facet (of 30), could be considered locatable within the Big Five. In Study 1, we found that ~90% of BIS scales and ~80% of BAS scales surpassed the 27th Big Five facet; we confirmed this pattern of findings in Study 2. While only 38% of FFFS scales surpassed the 27th facet in Study 1, most of these scales

(~70%) did so in Study 2. Overall, over 70% of RST scales were well explained by the Big Five in both of our studies, suggesting that the vast majority of these scales are plausible candidates for facets of the Big Five.

RQ2: Where can RST scales be located at the domain level?

Of the scales that could be located within the Big Five, two salient patterns emerged in both studies: All BIS scales were closely aligned with neuroticism, and most BAS scales were closely aligned with extraversion (though with some exceptions, as explored below). These findings broadly match the locations Gray (1970, 1981) specified for trait manifestations of the BIS and BAS within Eysenck's model (see Figure 1). Patterns of convergence between the BIS scales and neuroticism were particularly clear. Given that neuroticism characterises an individual's propensity to experience negative emotions, such as anxiety (Eysenck, 1967; John et al., 2008), for which the BIS is a putative basis, the strength of this relationship makes good theoretical sense. The locations of FFFS scales within the Big Five were somewhat more diffuse, as is described further below.

RQ3: Where can RST scales be located at the facet level?

Finally, several RST scales were highly correlated with specific Big Five facets. In both studies, we found that at least 75% of the included BIS scales were highly correlated with at least one facet of neuroticism, and all were strongly correlated with the Anxiety facet. The grounding of anxiety in BIS functioning is central to both o-RST and r-RST, and it seems that this strong theoretical backbone supported the development of several BIS scales with strong construct validity. In contrast, very few BAS and FFFS scales were highly correlated with existing facets.

In both studies, several strong correlations ($r > .80$) between RST scales and Big Five facets (predominantly between BIS scales and neuroticism facets), led us to ask whether any were sufficiently similar to be deemed *identical* (Lawson & Robins, 2021). By comparing single-factor to two-factor models, we found that differences in fit were marginal and came as a cost to parsimony, despite the fact that a two-factor model always offered significantly better fit. Thus, several RST scales appear to be so highly related to existing Big Five facets that they are approaching redundancy.

Key implications and recommendations

Overall, a clear majority of RST scales could be described within the space of the Big Five. This knowledge can help

to clarify similarities and differences among the plethora of RST scales available to researchers, and guide choices regarding scale use. Three broad sets of observations are particularly salient, and support several recommendations, as follows:

Close correspondence between BIS scales and neuroticism facets. Several BIS scales were highly correlated with existing neuroticism facets, with some verging on redundancy. For example, despite their slightly different labels, the RST-PQ BIS and the SPSRQ Sensitivity to Punishment scales were closely correlated (Tables 6 and 10) and had large latent variable correlations (exceeding $r = .60$) with the same three facets of neuroticism (Anxiety, Depression, and Self-Consciousness). Thus, it is reasonable to conclude that the RST-PQ BIS and SPSRQ Sensitivity to Punishment scales are measuring similar underlying constructs, and findings based on either scale can be integrated. Importantly, the fact that the Big Five facets predate these RST scales suggests that these RST scales may indeed provide examples of construct proliferation and jangle fallacies within personality psychology. Going forward, researchers should acknowledge the potential equivalences among these particular BIS scales and existing Big Five facet scales.

In stark contrast to all other BIS scales, Jackson-5 BIS was deemed peripheral to the Big Five in both of our studies. In our second study, this scale was most strongly aligned with extraversion (rather than neuroticism), which seems highly unexpected from the theory. This adds to Corr's (2016) concerns about the construct validity of the Jackson-5 BIS scale, given that it has been found to correlate more strongly with BAS scales compared to BIS scales or measures of trait anxiety, alongside its patently low face validity (e.g., items such as *I like my peers to know I'm doing well*). Overall, the available evidence suggests that—whatever construct the Jackson-5 BIS is measuring—it is (a) markedly *different* from that assessed by other prominent BIS scales, (b) not aligned with neuroticism/anxiety in the manner specified by Gray (see Figure 1), and (c) peripheral to the Big Five trait space altogether. Findings based on the Jackson-5 BIS scale may thus be misleading, and difficult to integrate with findings based on other BIS scales. Given the many alternatives available (e.g., other BIS scales, closely related Big Five facets, etc.), we recommend that the Jackson-5 BIS scale is simply not used in future research.

Divergent patterns for BAS scales. When designing the Carver-White BAS scale, the authors adopted a “divergent strategy, consistent with the absence of complete consensus about how exactly BAS sensitivity is likely to manifest” (Carver & White, 1994, p. 322). Thirty years later, that lack of consensus is still evident across the

various BAS scales. Along with strong associations with extraversion, two contradictory patterns emerged involving associations with either high conscientiousness (e.g., RST-PQ Goal-Drive Persistence) or low conscientiousness (e.g., RST-PQ Impulsivity, Carver-White Fun Seeking).⁴ Although neither o-RST nor r-RST explicitly forecasts either of these associations, they can nevertheless be reconciled with theory. Specifically, two alternative conceptualisations of the trait output of the BAS have been termed *rash impulsiveness* (arguably the conceptualisation within o-RST) and *reward drive* (a reconceptualization of trait BAS proposed alongside r-RST; Dawe et al., 2004; Smillie et al., 2006). Rash impulsiveness aligns with common conceptualisations of impulsivity as a tendency toward disinhibited behaviour, engaged with little planning or forethought, which is indicative of low conscientiousness (e.g., Whiteside & Lynam, 2001). Meanwhile, reward drive aligns with conceptions of goal-directedness and self-regulation, which are indicative of high conscientiousness (e.g., Corr et al., 2013). For some time, many have argued that reward drive is a more theoretically plausible conceptualisation of the BAS-related trait than rash impulsiveness (see Dawe et al., 2004; Depue & Collins, 1999; Pickering, 2004; Smillie et al., 2006). Indeed, compared to BAS scales reflecting reward drive (e.g., Carver-White Drive scale), some evidence suggests that BAS scales reflecting rash impulsiveness are not associated with reward-contingent learning (Dawe et al., 2004; Pickering, 2004; Smillie & Jackson, 2006). Drawing these points together, we suggest that BAS scales aligned with lower conscientiousness may reflect a somewhat outdated conceptualisation based on rash impulsiveness, and recommend that researchers favour BAS scales aligned with higher conscientiousness, though primarily with extraversion.

Beyond these patterns of divergence, it should also be noted that some BAS scales were unable to be consistently located within the Big Five across our two studies. The most salient example of this was the SPSRQ Sensitivity to Reward scale, which reflected the Big Five to an average degree in Study 1 (>20th facet) but was deemed peripheral in Study 2 (<27th facet). Although we can only speculate, we note that our sample for Study 2 was older and more gender-balanced compared to that of Study 1. Given that Conner et al.'s (2018) SPSRQ was validated on an undergraduate sample that greatly resembled the Study 1 sample, it is possible that the inconsistent results may reflect poor generalisability from a student sample to the community sample used in Study 2. Whatever the explanation, our findings relating to SPSRQ Sensitivity to Reward are inconclusive.

Fragmented patterns for FFFS scales. Compared to the clear patterns observed for the BIS scales, and to a certain degree the BAS scales, observed patterns for the FFFS

scales were much murkier. Whilst associations with high neuroticism and low extraversion typically characterised the Freeze scales, the Reuter-Montag Freeze scale, had a primary, negative relationship with conscientiousness. This may indicate that Freeze sensitivity measured by the Reuter-Montag scale inadvertently taps into disengagement and lack of persistence—that is, freezing (i.e., doing nothing) in response to a threat may be less conceivable to individuals higher on conscientiousness. Second, associations the Reuter-Montag Fight scale had with high extraversion, low agreeableness, and low neuroticism, may indicate possible contamination with predatory aggression. As Fight scales should reflect *defensive* aggression, we would generally expect them to have positive associations with neuroticism (Corr et al., 2013). Given that Reuter et al. (2015) also found their Fight scale to be negatively associated with their total FFFS scale, it seems possible that it may be measuring a more antagonistic trait concept. We therefore encourage readers to exercise caution when using Fight scales as a measure of defensive aggression, as some may inadvertently assess predatory aggression.

It is also important to note that most FFFS scales were considered to lie outside of the Big Five. Indeed, of the eight FFFS scales included in both studies, at least 50% were deemed to be peripheral to the Big Five in both studies. This may be owing to several issues that are likely to make self-report measurement of FFFS constructs difficult. Perhaps the most obvious is the low base rate of Fight, Flight, and Freeze in everyday life: Unlike the rodent participants described by Gray and McNaughton (2000, chapter 2) most human participants rarely encounter inescapable threats, natural predators, or other stimuli thought to engage the FFFS. This may make it difficult to introspect upon one’s own FFFS sensitivity (Corr et al., 2013; Reuter et al., 2015; Smillie et al., 2011), perhaps explaining the fragmented and inconsistent links between FFFS scales and broad dimensions of personality. If so, then our findings underscore the potential value of supplementing self-report scales with behavioural measures (Corr, 2013)— such as computer simulated exposure paradigms (see Mobbs et al., 2007), to measure Fight, Flight and Freeze responses—rather than self-reports alone.

To summarise, Table 12 presents the key recommendations we would offer researchers in this area based on our findings. We also call for a moratorium to be placed on the development of further RST questionnaires given the “muddled” confusion that has resulted from four decades of unabated scale proliferation in this area (Corr, 2016, p. 60).

Limitations and directions for future research

The R^2 criteria we established for the 10th, 20th, and 27th facets in both of our studies are considerably lower than those established by Bainbridge et al. (2022). Indeed, if the criteria derived by Bainbridge et al. (2022) for the 27th facet (R^2 of .43) was used in the current research, only 10 scales (37%) in Study 1 and 16 scales (48%) in Study 2 would be deemed locatable within the Big Five. Differences between our criteria and those derived by Bainbridge and colleagues may have arisen for several reasons. First, Bainbridge et al. (2022) analysed meta-analytic R^2 values across three samples, which may have yielded more robust estimates compared to the individual samples used in the current paper. Thus, it is possible that the criteria in the current study were underestimated. Alternatively, it is possible that our rigorous quality control checks (as detailed in Supplemental section A) reduced the impact of socially desirable responding and other biases that could inflate correlations among self-reports. In this case, our more conservative estimates may be more reliable than those reported by Bainbridge et al. (2022). Regardless of the explanation for these discrepant estimates, it is important to bear in mind that the method devised by Bainbridge et al. (2022) and employed here enable each sample to act as its own control. This is because any factors that attenuate (or inflate) correlations between the Big Five domains and their facets would likely also attenuate (or inflate) correlations between the Big Five domains and RST scales (see Bainbridge et al., 2022, for further explanation). Finally, it is worth noting that some additional analyses conducted by Bainbridge et al. (2022) were not implemented in the current study. The more streamlined analysis adopted in the current work does come with non-zero risks of some misleading results. In saying this, it is unlikely that these risks

Table 12. Summary of recommendations for RST scales.

System	Recommendation
BIS	Most BIS scales can be used, perhaps even interchangeably, but avoid the Jackson-5 BIS scale. Many facets of neuroticism (in particular, anxiety) may work just as well as a purpose-built BIS scale.
BAS	Favour scales associated with high extraversion and high conscientiousness. Avoid scales associated with low contentiousness, and regard the SPSRQ sensitivity to reward scale with caution. Some facets of extraversion (e.g., excitement seeking) may be good substitutes for a purpose-built BAS scale.
FFFS	Regard all scales with caution, especially fight scales, which may inadvertently measure predatory (rather than defensive) aggression.

Note. BIS = Behavioural Inhibition System. BAS = Behavioural Activation System. FFFS = Fight-Flight-Freeze System.

alter our most salient findings: that a vast majority of BIS scales overlapped highly with the Big Five and were primarily aligned with neuroticism. For further details of the differences between the analyses completed in the two studies, please see [Supplemental section A](#).

We acknowledge that our use of convenience samples may reduce the generalisability of the present findings. This is of particular pertinence to RST research as both trait anxiety (BIS) and reward dysregulation (BAS) are topics of relevance to clinical populations, including those with eating disorders (e.g., [Wilson et al., 2019](#)), anxiety disorders (see [Afshari, 2020](#)), and substance use disorders (e.g., [Boog et al., 2013](#)). It is obviously not safe to assume that results from an undergraduate university sample (Study 1), or a community sample (Study 2), will generalise readily to such clinical samples. Future work should therefore prioritise replication in these communities. Importantly, however, our results were broadly similar across the two very different samples employed here, suggesting that results are likely generalisable in non-clinical samples.

Relatedly, we call for extensions of the current study to examine additional RST scales (see [Table 1](#)) in order to determine whether our conclusions generalise to scales we did not examine here. This will allow for a more comprehensive appraisal of the state of RST measurement, above and beyond what can be achieved with the inclusion of only five RST questionnaires. Indeed, if a robust picture of the state of RST measurement could be created, efforts could be made to clarify the measurement space. Thus, the trajectory of replication efforts should focus on rejecting scales with low construct validity and synthesising similar scales into a single comprehensive RST questionnaire.

Finally, although the current study may help to constrain and organise self-report measures of RST constructs, it also important to acknowledge that self-report scales alone are unlikely to be sufficient for measuring the sensitivities of the biological systems proposed in RST (see [Smillie, 2008](#)). The reliance on questionnaires to infer the sensitivity of underlying biological systems involves linking self-report to physiological data – two distally related units of analysis ([Patrick & Hajcak, 2016](#)) that only show modest covariation ([Patrick et al., 2013](#)). Along the lines proposed in the Research Domain Criteria (RDoC) framework ([Insel et al., 2010](#)), we argue that future research in this space should assess RST systems at different levels of measurement beyond solely self-report. Applying a multi-modal approach to RST research will help to triangulate on the higher-order systems proposed in the theory. For example, researchers seeking to study the BAS may apply relevant self-report scales identified in the current study (e.g., Excitement Seeking facet of extraversion or the RST-PQ BAS scale) in tandem with behavioural paradigms, such as the Effort Expenditure for

Reward Task ([Treadway et al., 2009](#)), to more comprehensively study the system. In sum, a multi-modal approach should be considered in future research that seeks to examine the links between personality traits and the variation in the biological systems proposed in RST.

Conclusion

As more and more RST questionnaires have been developed, it has become increasingly difficult for researchers to navigate and synthesise the empirical literature in this area. In the present research, we found that most of a large collection of RST scales were locatable within the Big Five framework and are plausible candidates for facets of the Big Five domains. Furthermore, some of these RST scales overlapped with existing Big Five facets to such a degree as to approach redundancy with those facets. By contextualising RST scales in Big Five framework, the present research helps address [Corr's \(2001, 2016\)](#) previously discussed concerns by proposing an organising framework that (a) reduces confusion regarding the psychometric properties of RST scales, and (b) offers a rationale for choosing between scales. By mapping these various scales within the Big Five trait space, the present findings, and future extensions of this work, may help RST researchers to integrate and build upon findings in this area.

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Author contributions

Allyson. K. Alblas: Data curation; Formal analysis; Methodology; Project administration; Validation; Visualization; Writing – original draft; and Writing – review & editing.
Timothy. F. Bainbridge: Conceptualization; Formal analysis; Methodology; Visualization; and Writing – review & editing.
Luke. D. Smillie: Conceptualization; Data curation; Formal analysis; Methodology; Project administration; Supervision; Validation; Visualization; Writing – review & editing.

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Data accessibility statement

The data needed to reproduce the study results are available on the OSF at <https://osf.io/q9vsk/>. This link offers open access to all data for analysis, pre-registrations, R script, and a codebook with additional information.

Supplemental material

Supplemental material for this article is available online. Depending on the article type, these usually include a Transparency Checklist, a Transparent Peer Review File, and optional materials from the authors.

Notes

1. A further recent revision by [McNaughton and Gray \(2024\)](#), published just after our studies were completed, will not be discussed in this paper.
2. The one exception was Liberalism facet of Openness, $r = .19$. As noted by [Bainbridge et al. \(2022\)](#), some of the Liberalism items (e.g., *Tends to vote for liberal political candidates*) may be misinterpreted in an Australian sample, given that the major conservative party is called the Liberal Party.
3. This was quite a stringent test given that these correlations fell just below this threshold in Study 1.
4. A third, more subtle pattern involved associations with low agreeableness (e.g., SPSRQ sensitivity to reward). This may reflect potential links between reward motivation and heightened predatory aggression required to obtain reward ([Mitchell et al., 2007](#)), but we caution that this result was less robust across the two studies.

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